

DANIELE PESSINA

+44 7426363133 | danielepessina99@gmail.com | [linkedin.com/in/pessinadaniele](https://www.linkedin.com/in/pessinadaniele) | danielepessina.github.io/

EDUCATION

Imperial College London

2022 – Present

PhD in Chemical Engineering - CASE Studentship with AstraZeneca

- Thesis title: A hybrid *in-silico/in-vitro* approach for integrated design and optimisation of large-molecule crystallisation

Imperial College London

2018 – 2022

MEng in Chemical Engineering, 1st Class Honours

EXPERIENCE

PhD Researcher, Process Systems Engineering

2022 - Present

Supervision: Dr. Maria Papathanasiou & Prof. Jerry Y. Y. Heng

- Developed novel methods for process simulation, optimisation and analysis of crystallisation and pharmacokinetic systems governed by coupled ODE/PDE dynamics, including both steady-state and dynamic modelling. Designed physics-informed ML architectures (neural ODEs, neural operators, transformers) as surrogate models and digital twins for high-fidelity process simulation under thermodynamic and transport constraints, validated against gPROMS and Aspen Plus process models.
- Built hybrid mechanistic/ML models combining first-principles process equations with neural networks, enabling transfer learning across sparse, multi-fidelity datasets and new operating conditions. Applied neuro-symbolic methods for mechanistic interpretability and reduced model complexity, supporting transparent process understanding.
- Built GPU-accelerated, autodifferentiable probabilistic inference pipelines in JAX and Julia for uncertainty-aware parameter estimation, robust process optimisation under uncertainty and global sensitivity analysis of large-scale dynamical process models. Optimised computations through JIT compilation, adjoint methods, and multi-thread/multi-node parallelism on HPC clusters.
- Developed agentic AI coding workflows to autonomously generate, test and iterate on research code, accelerating experiment turnaround and reducing manual implementation overhead.
- Developed ML-enhanced global sensitivity analysis and model reduction methods for complex, intractable pharmaceutical process models, enabling identification of critical process parameters and simplification of high-dimensional systems.
- Conducted *in-vitro* medium-scale lysozyme crystallisation experiments with PAT tools, generating experimental evidence for template-induced protein nucleation mechanisms and closing the loop between simulation predictions and laboratory validation.

Data Scientist

Jan. 2026 - Present

Quaisr, London

- Deployed end-to-end agentic AI workflows for pharmaceutical and life sciences clients in production, embedding directly with customer teams to digitise and automate core process development workflows from experimental design through optimisation to deployment.
- Integrated external engineering tools, simulators and client data systems into automated pipelines via APIs and custom wrappers, enabling AI agents to autonomously plan, execute and iterate on multi-step process engineering tasks.

Consultant Data Scientist

March - July 2025

Haleon

- Delivered ML-based soft sensors for real-time product quality estimation on manufacturing floors, from large-scale, high-frequency manufacturing data, reducing reliance on offline batch testing and integrating predictions into production monitoring workflows.
- Benchmarked LSTM and transformer models against classical feature-engineered approaches on industrial process datasets, optimising for robustness and accuracy under distribution shift and imbalanced data regimes.

Lixea – Summer Intern

June - Sep. 2021

Supervision: Prof. Jason Hallett

- Developed a techno-economic analysis (TEA) model for large-scale biomass fractionation processes using Aspen Plus, quantifying process uncertainty, capital cost variability and operating expenditure sensitivity. Provided strategic forecasts directly used in investor reports.

TECHNICAL SKILLS

Core: Platform deployment and system integration, workflow automation, agentic AI systems, digital twins, process simulation and optimisation, physics-informed ML, neural operators, ODE/PDE numerical modelling, surrogate modelling, Bayesian inference, uncertainty quantification, global sensitivity analysis

Software: Python, JAX, PyTorch, Julia, MATLAB, R, Git, Docker, Azure, Databricks, MLflow, PySpark, SQL

Process tools: gPROMS, Aspen Plus, LCA

HPC: SLURM, multi-node parallelism, JIT compilation, GPU-accelerated MCMC and variational inference, adjoint sensitivity methods

PUBLICATIONS

Heng, J. et al. (2026) Biomolecular Crystallisation Through Soft Templates and Seeding; In: *Advances in Biochemical Engineering/Biotechnology*, Springer

Pessina, D. et al. (2026) Transfer Learning of Data-driven Crystallisation Processes via Constrained Neural Ordinary Differential Equations; *Dig. Chem. Eng.*

Pessina, D. et al. (2025) Model-based approach to template-induced macromolecule crystallisation; *Systems & Control Transactions*

Pessina, D. et al. (2025) Machine learning-enhanced Sensitivity Analysis for Complex Pharmaceutical Systems; *Systems & Control Transactions*

Pessina, D. et al. (2025) Integrated In Vitro/In Silico Uncertainty Quantification Method for Protein Crystallization Models; *Industrial & Engineering Chemistry Research*

OPEN-SOURCE CONTRIBUTIONS

CriSTool | *Julia*-based simulation platform for crystallisation population balance models with uncertainty quantification

gsax | JAX-accelerated package for fast global sensitivity analysis of large-scale dynamic models.